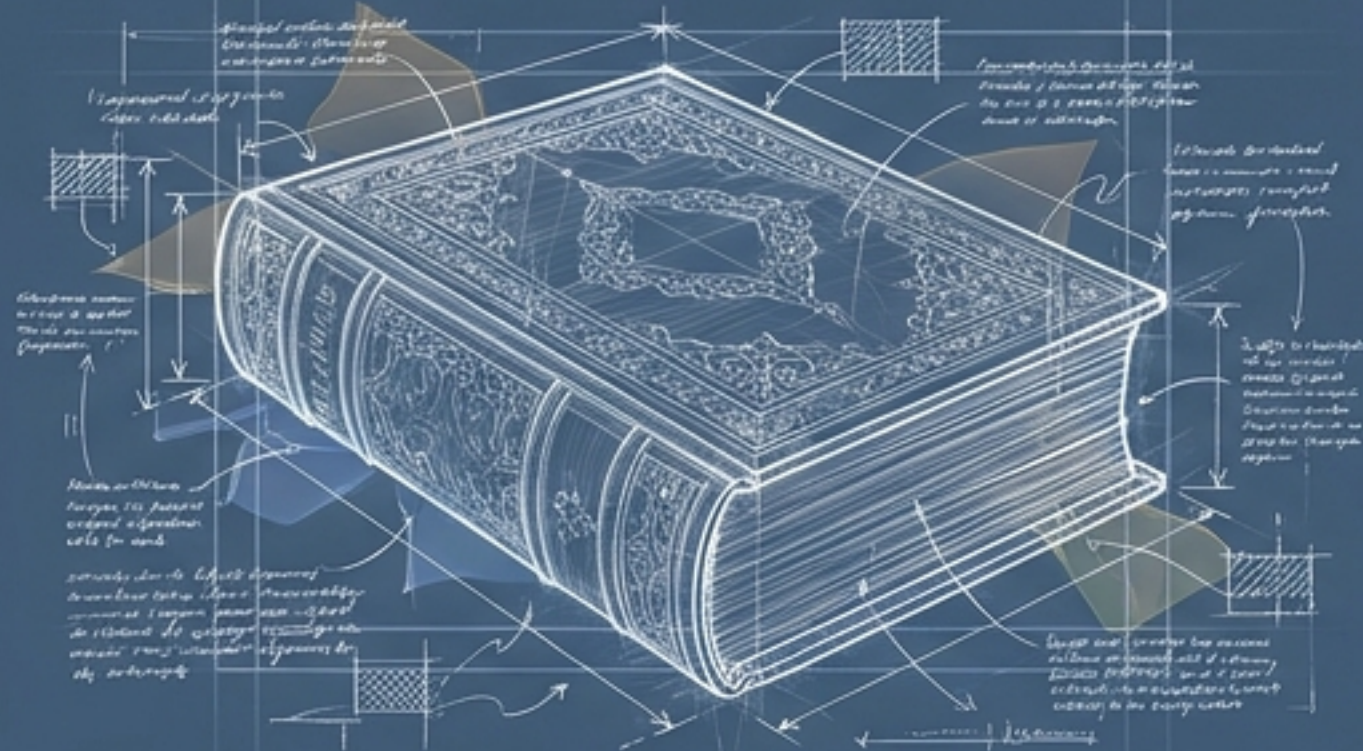




Leaving out detail is a feature, not a flaw

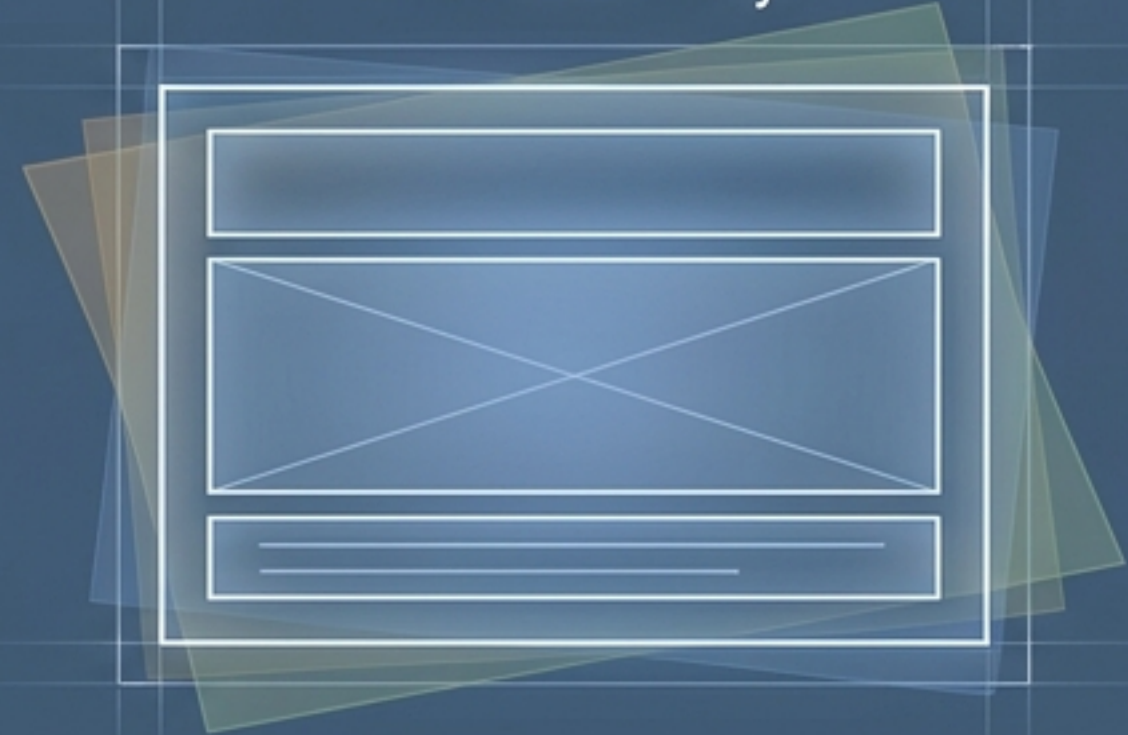
Italian Translation



Representation

Attempts to maintain all original information word-for-word.

Slide Summary



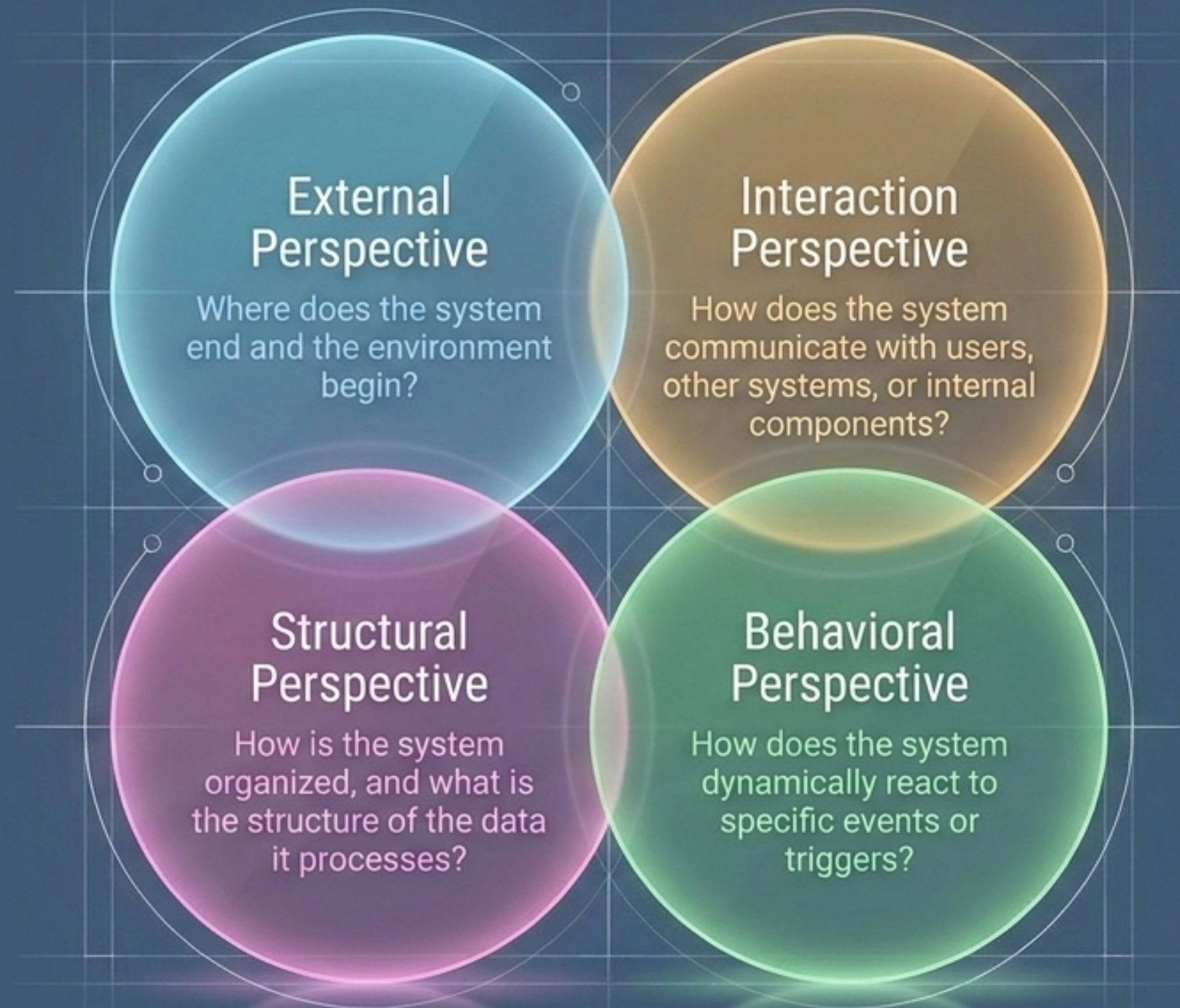
Abstraction

Deliberately simplifies design to highlight the most salient characteristics without the weight of every sentence.

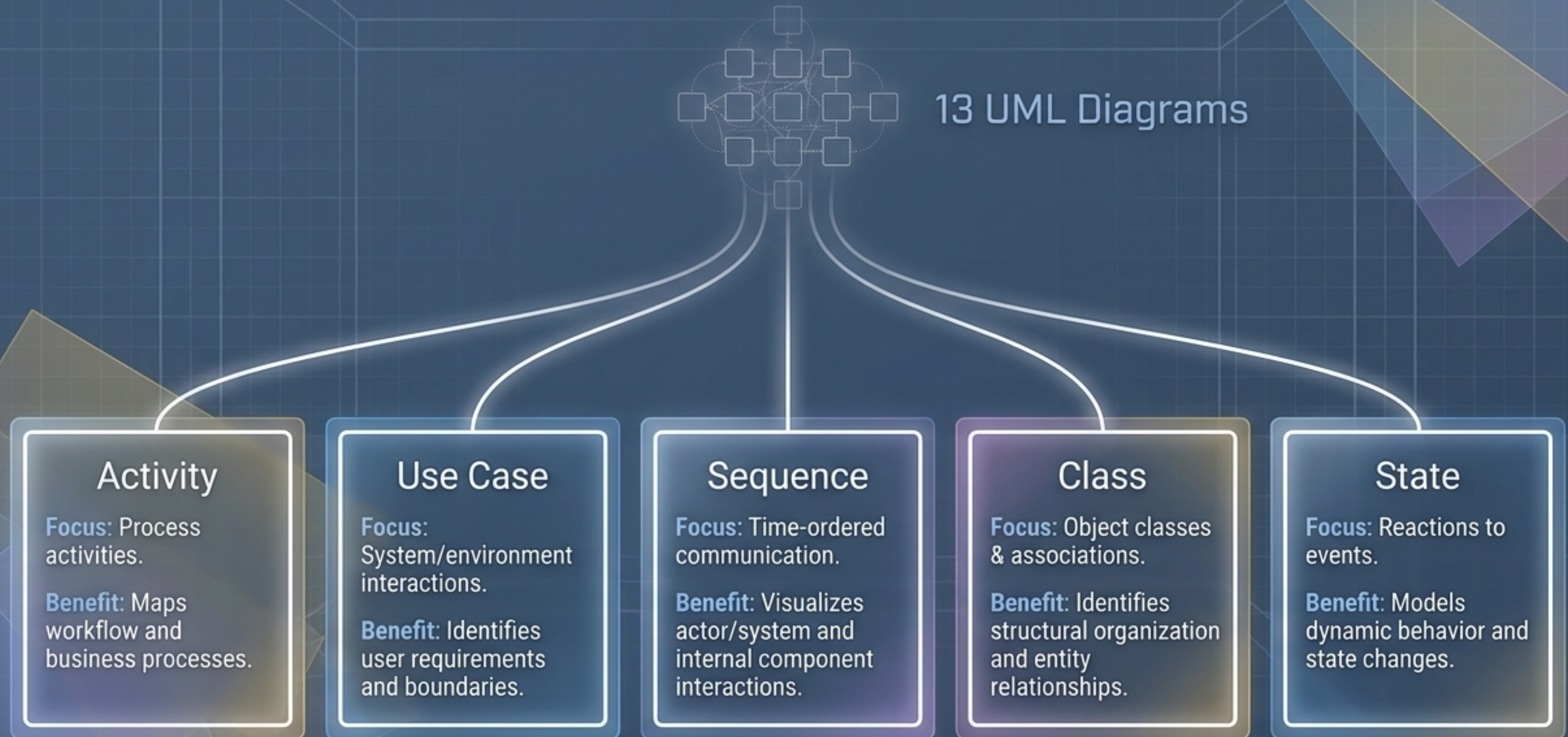
Senior Architect Insight:

Students often worry a model is flawed if it lacks full system detail. In reality, systematic abstraction is the primary goal of the Unified Modeling Language (UML).

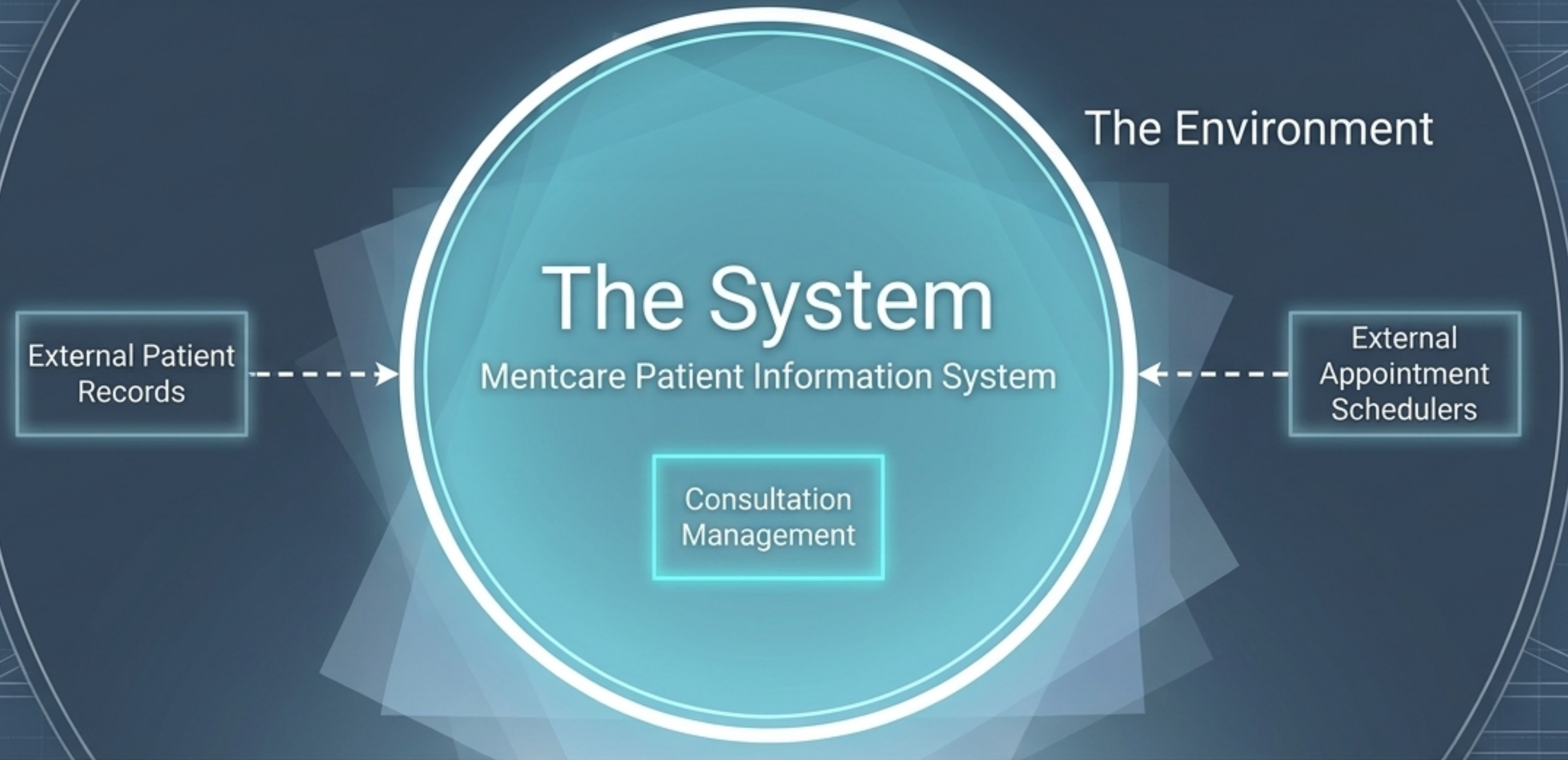
Four distinct lenses for viewing a single system



Distilling 13 UML standards down to the essential five



Defining the system context boundary

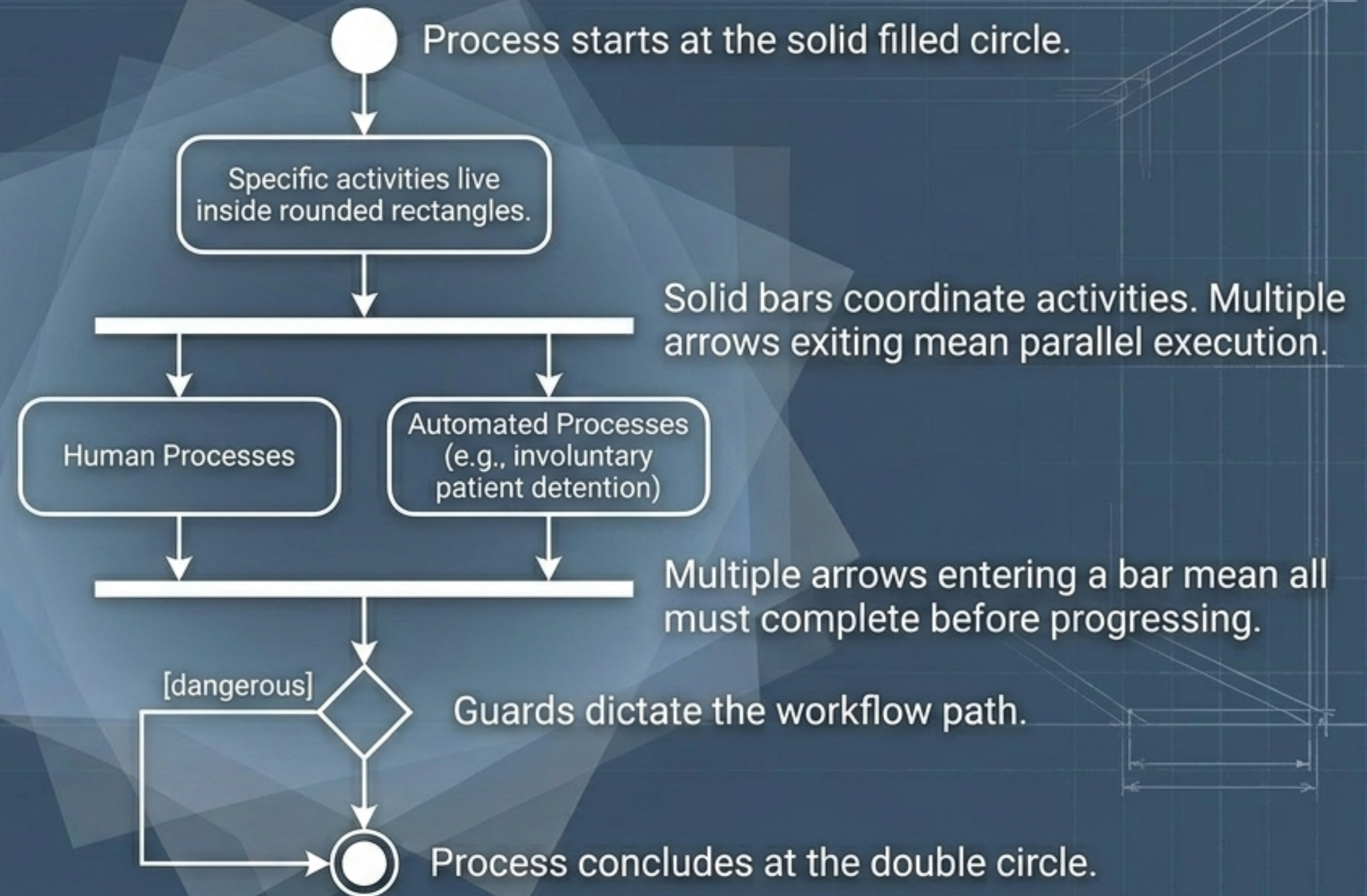


Early in a project, you must decide what functionality stays inside the software and what remains part of the external environment.

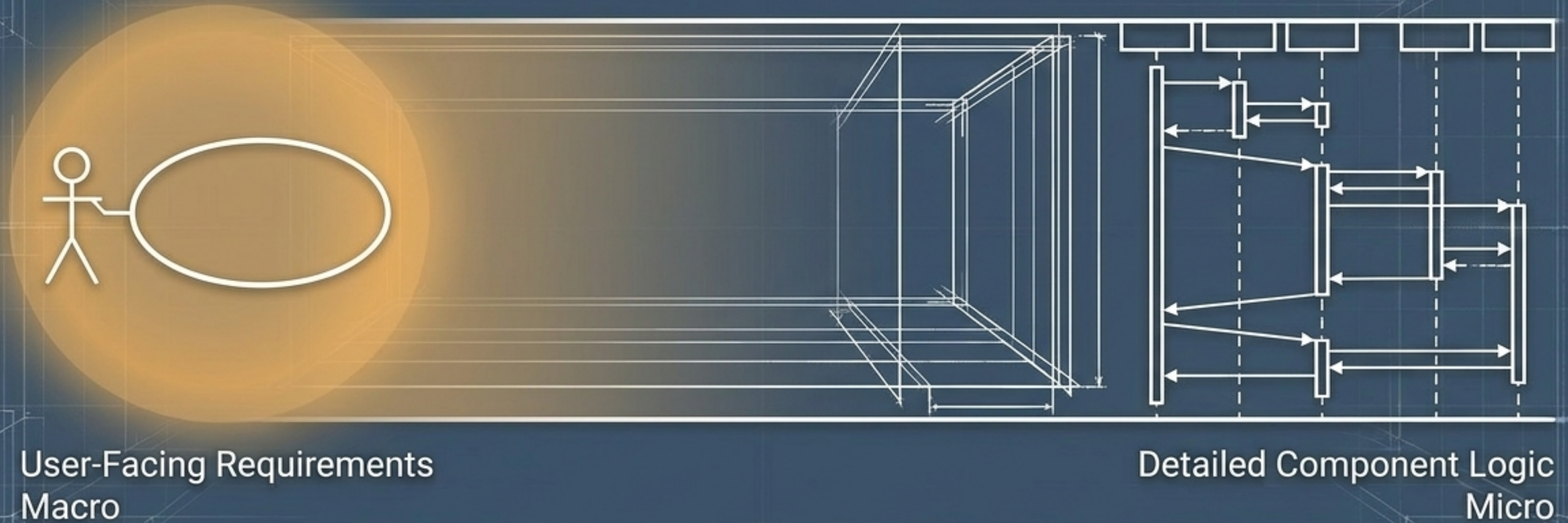
Non-technical forces shape technical boundaries



Modeling business processes with Activity Diagrams



Interaction modeling spans from macro requirements to micro component logic

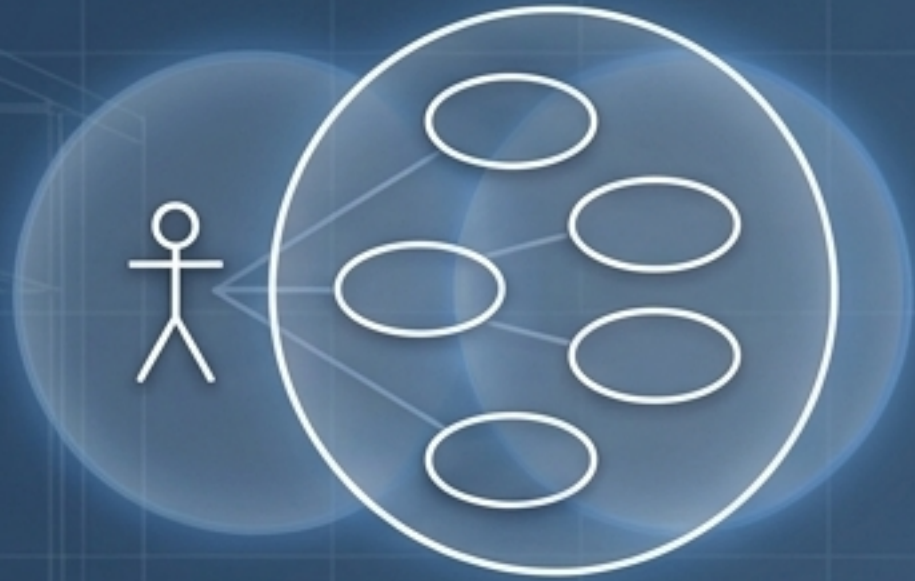


The Communication Diagram Caveat:

While standard UML includes 13 diagrams, we prioritize Sequence over Communication diagrams. Communication models are simply an alternative representation of the same sequence information.

Choosing your interaction tool: Use Case vs. Sequence

Use Case Diagrams

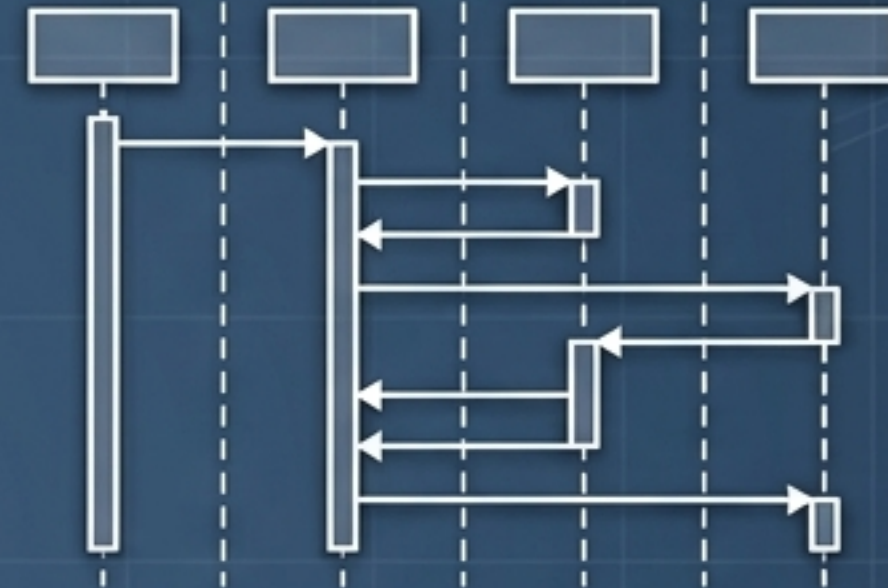


Level of Detail
High-level overview of goals.

Primary Actors
External agents only (humans or other systems).

Origin/Standard
Developed by Ivar Jacobsen (1993) to capture user requirements.

Sequence Diagrams



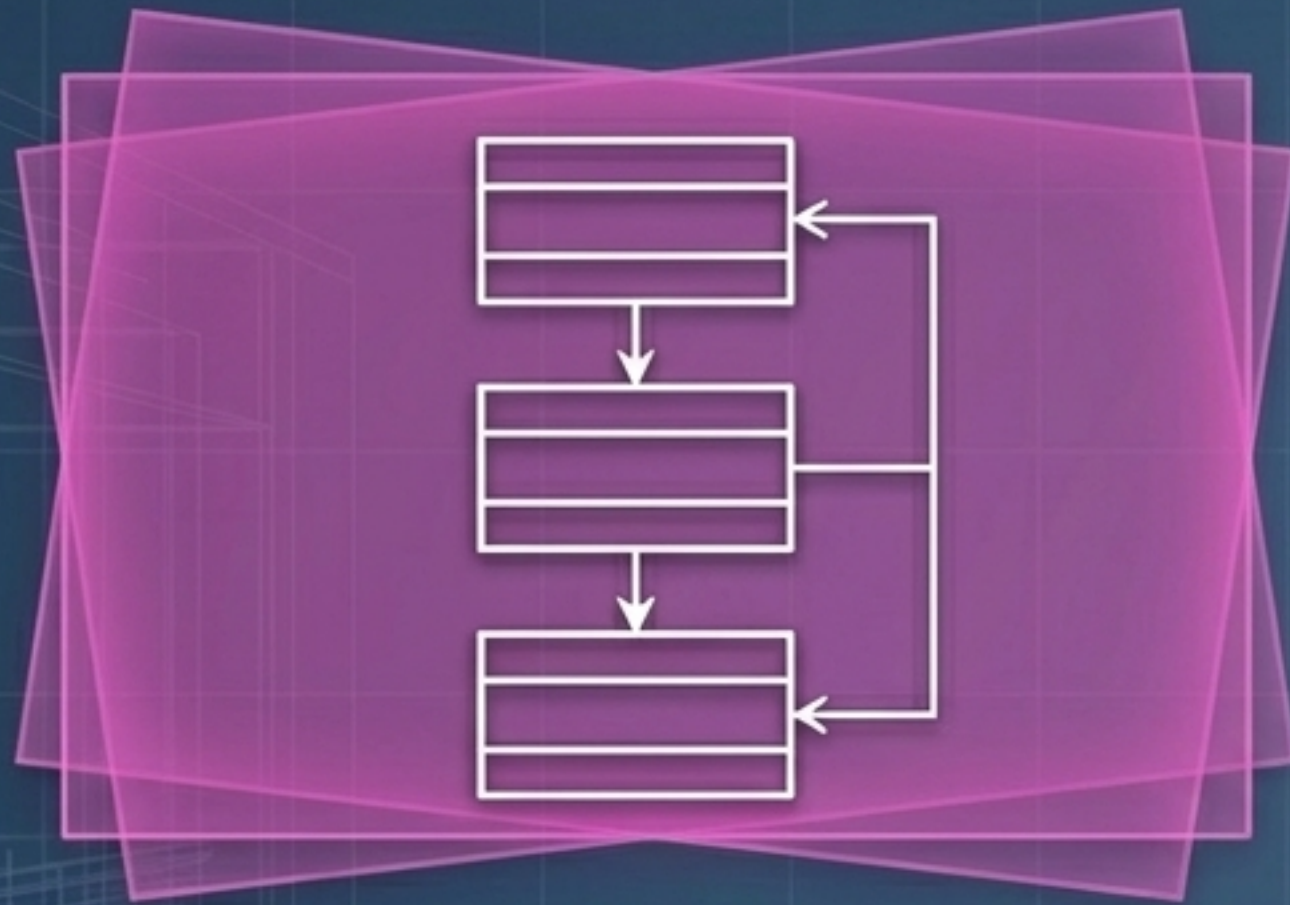
Level of Detail
Granular, step-by-step logic.

Primary Actors
Both external agents and internal components.

Origin/Standard
Standard UML tool for documenting precise internal logic.

Form follows function: The purpose of your model dictates the required detail.

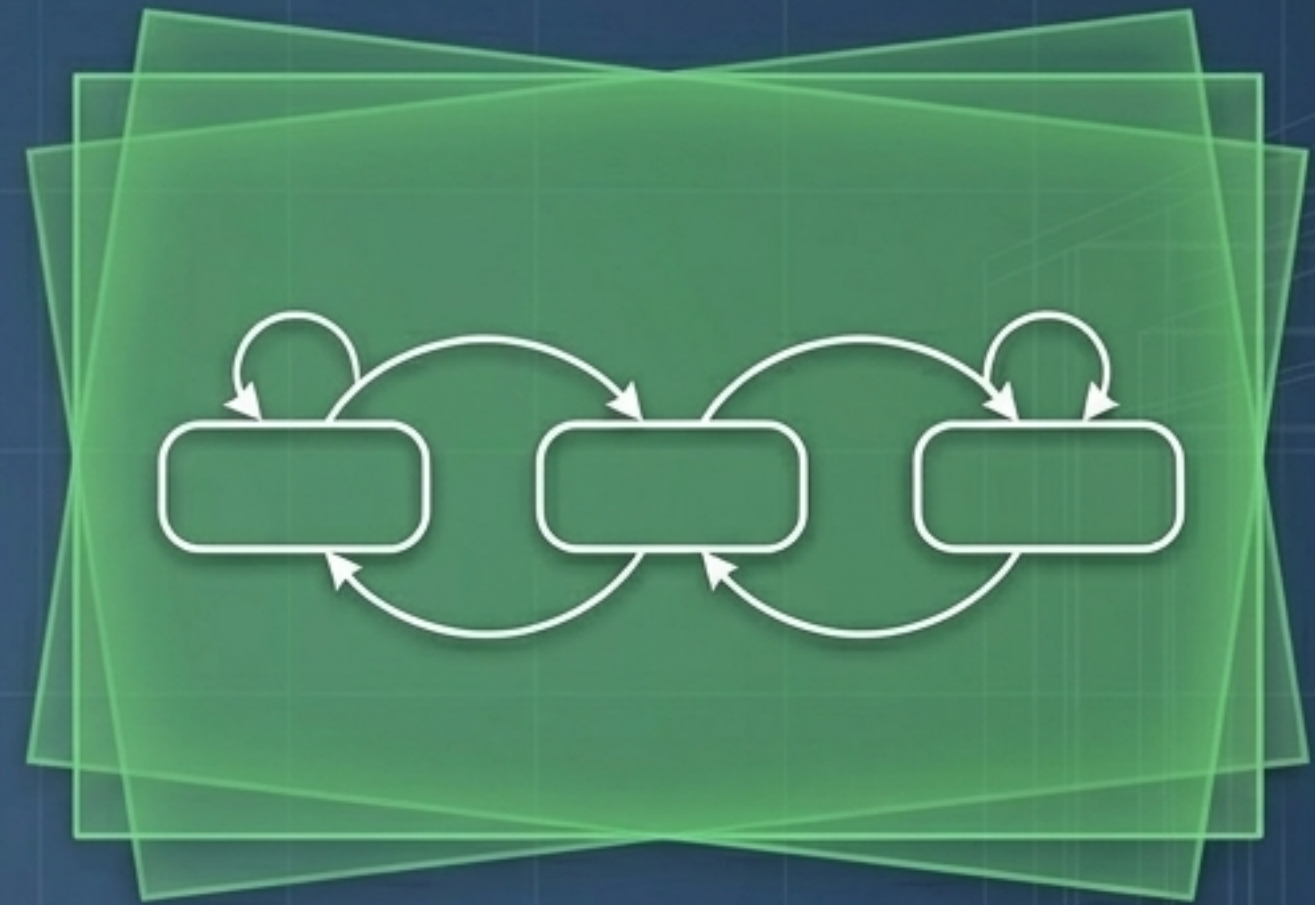
Defining structure and state dynamics



Class Diagrams

Structural Perspective

Identifies the structural organization of object classes and the fixed associations between data entities.

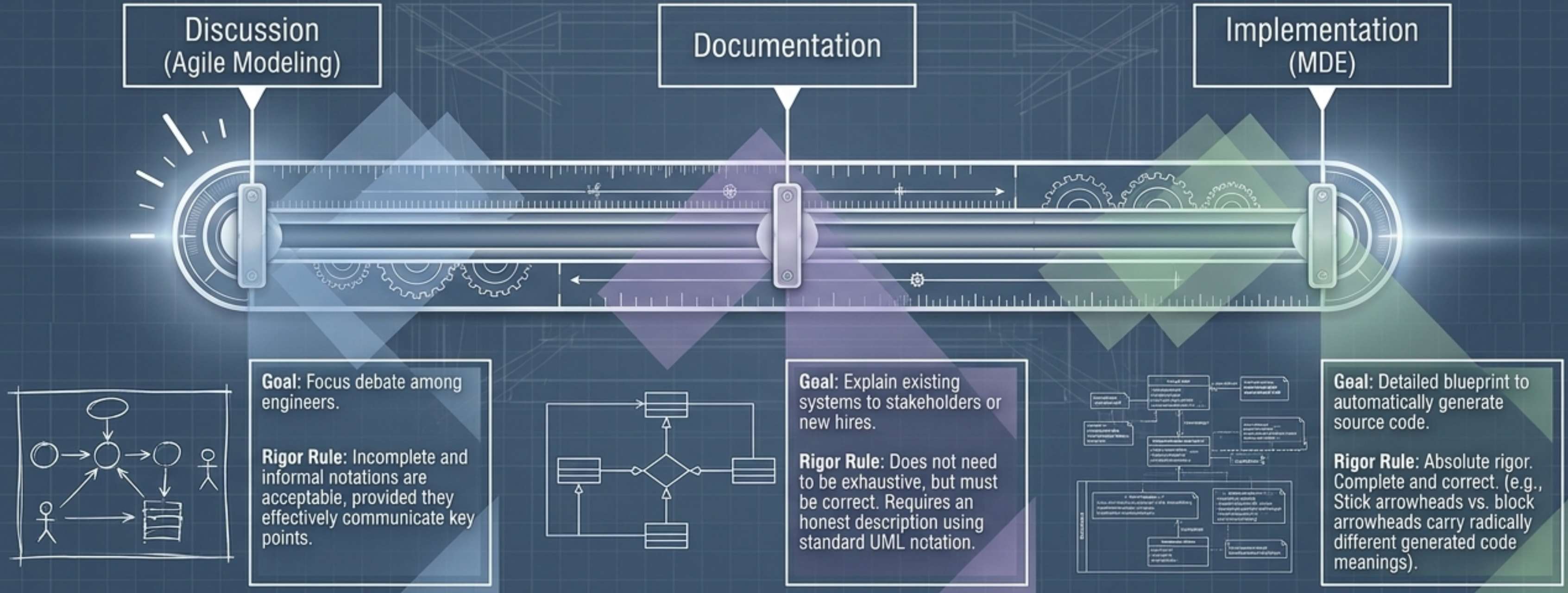


State Diagrams

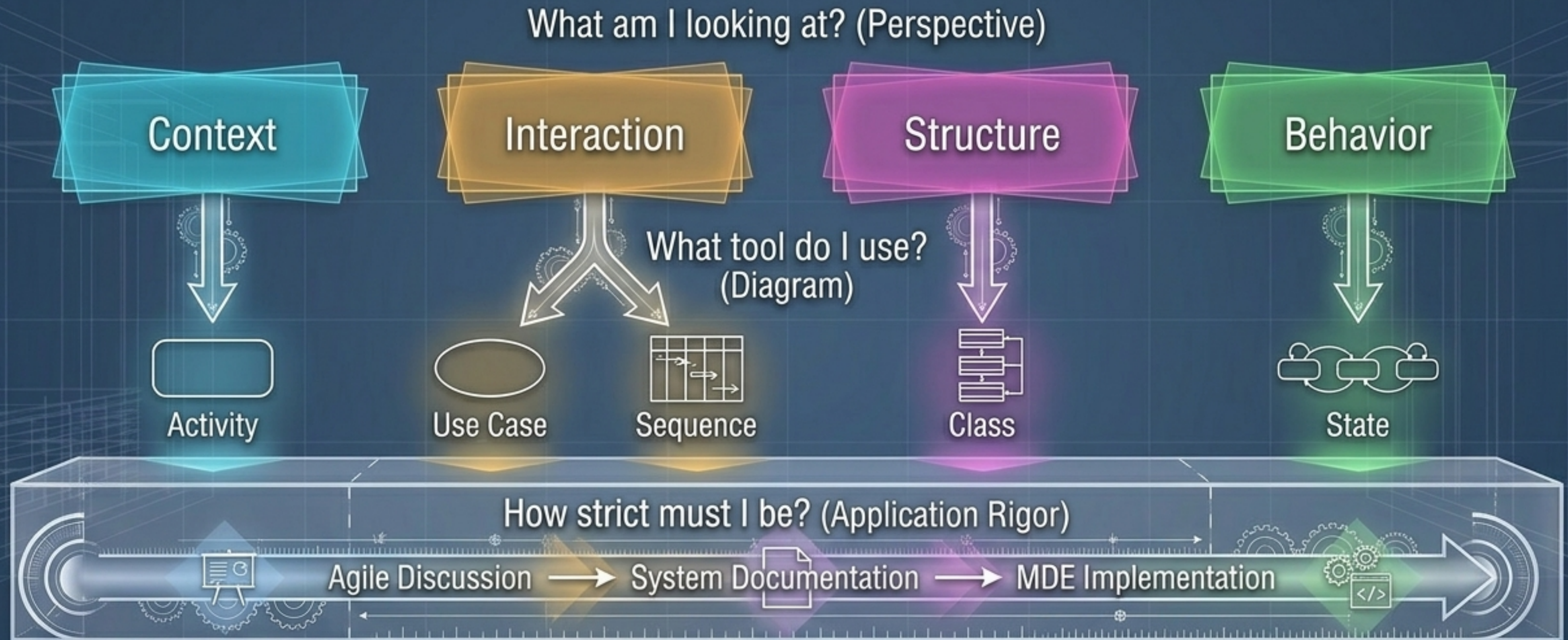
Behavioral Perspective

Models how a system dynamically changes state in direct response to internal and external events.

The Application Spectrum: Three levels of technical rigor



The Unified System Modeling Taxonomy



System modeling is not about memorizing symbols. It is a flexible discipline of matching the right perspective, to the right diagram, with the necessary rigor.